

AA-PERPHENAZINE

Perphenazine Tablets USP 4mg

Antipsychotic-Antiemetic-Antianxiety

Pharmacology: Perphenazine is a piperazine phenothiazine derivative. Numerous phenothiazine derivatives are in current use. They resemble chlorpromazine in action but differ from it in potency, clinical indications, and toxicity.

The pharmacologic effects of the phenothiazine tranquilizers are quite complex. In addition to their behavioral effects, these drugs are potent anti-emetics and have important actions on the autonomic nervous system at various levels. In large doses, they also produce significant toxic side effects such as parkinsonism.

The various phenothiazines differ from the standpoint of their potency, extrapyramidal effects, and sedative effects. The piperazines are more potent than other phenothiazines, cause significant extrapyramidal effects, and are not very sedative.

Since the extrapyramidal effects may be antagonized by the anticholinergic drugs, it is probable that differences in causing extrapyramidal side effects reflect intrinsic anticholinergic potency of some of the phenothiazines.

Chlorpromazine will be discussed as the prototype of the phenothiazine drugs.

Chlorpromazine an aliphatic or dimethylaminopropyl phenothiazine, is a sedative antipsychotic drug. In addition to its usefulness in agitated psychotic persons, the drug has other medical applications such as prevention of nausea and vomiting. It is used also for its ability to potentiate the actions of other drugs such as anesthetics.

The pharmacologic effects of chlorpromazine and other phenothiazines are antipsychotic, abnormal movements, anticholinergic and antiadrenergic, interference with temperature regulation, antiemetic, and endocrine.

Chlorpromazine, the prototype phenothiazine, is absorbed somewhat erratically when given by mouth. Peak plasma levels occur in 2 to 4 hours, a plateau is maintained for several hours, and predose levels occur in 12 hours. Plasma half-lives are extremely variable, averaging 9 hours.

The distribution of chlorpromazine in various tissues is quite uneven. The brain contains about four times as much as the plasma. Elimination takes place through the kidney and bile. The drug is excreted also in the milk. Chlorpromazine has numerous metabolites. The glucuronides and sulfoxides are excreted in the urine, and the demethylated, nonpolar metabolites enter the bile and may undergo reabsorption, excretion in the feces, or storage in the tissues.

Indications: Perphenazine, a piperazine phenothiazine derivative, is indicated in the treatment of psychotic disorders particularly to produce a quieting effect in hyperactive or excited psychotic patients. Used in the treatment of behavioral disorders in adults, in the aged and in children, and to prevent or relieve nausea and vomiting due to stimulation of chemoreceptor trigger zone.

Contraindications: Comatose or depressed states due to CNS depressants; blood dyscrasias; bone marrow depression; liver damage. Hypersensitivity to perphenazine; cross-sensitivity to other phenothiazines may occur.

Precautions: Hypotension, which is typically orthostatic, may occur especially in elderly and in alcoholic patients. This effect may be additive with other agents that cause a lowering of blood pressure. Exercise special care in those patients in whom a hypotensive crisis would be undesirable such as those with arteriosclerosis or other cardiovascular disease.

Prolongation of the QT interval, flattening and inversion of the T wave and appearance of a wave tentatively identified as a bifid T or a U wave have been observed in some patients receiving phenothiazines. These changes appear to be reversible and related to a disturbance in repolarization. Give phenothiazines cautiously to patients with heart disease.

Most reported cases of agranulocytosis associated with the administration of phenothiazine derivatives have occurred between the fourth and tenth week of treatment. Therefore observe patients on prolonged therapy with particular care during that time for the appearance of such signs as sore throat, fever and weakness. If these symptoms appear discontinue the drug and perform WBC and differential counts. Routine periodic WBC and differential counts are also advisable during prolonged but uncomplicated therapy.

If bilirubinemia, bilirubinuria or icterus occur, discontinue the drug and perform liver function tests. Routine periodic liver function tests are advisable during prolonged therapy. Phenothiazines have been associated with retinopathy. Discontinue perphenazine if retinal changes are observed. Use perphenazine cautiously in patients with a history of seizures. The occasional increase in physical activity resulting from perphenazine administration may augment the severity of pain in angina pectoris patients. Observe such patients carefully and withdraw the drug if necessary. Perphenazine's antiemetic effect can obscure signs of toxicity due to overdosage of the other drugs, or mask the symptoms of disease.

Contact dermatitis has been reported with a perphenazine solution; therefore, contact of hands or clothing by those handling perphenazine solutions should be avoided.

Occupational hazards: Where patients are participating in activities requiring complete mental alertness such as driving an automobile or operating machinery, administer the phenothiazine cautiously, forewarn the patient and increase the dosage gradually.

Pregnancy: Safe use of phenothiazines in pregnancy has not been established. Most studies indicate these agents are not teratogenic but there are reports of defects in infants exposed to these drugs during the first trimester. Toxic effects observed after high doses near term include: hypotonia, lithargy, depressed reflexes, paralytic ileus, jaundice, and persistent extrapyramidal syndrome. Therefore, they should be administered cautiously to women of childbearing potential particularly during the first trimester of pregnancy and near term.

Lactation: Phenothiazines are distributed into milk. Use with caution during lactation because of the possible sedative and anticholinergic side effects to the infant.

Drug Interactions: Phenothiazines may increase the effects of general anesthetics, opiates, barbiturates, alcohol and other CNS depressants as well as atropine and phosphorus insecticides. Perphenazine may block the antihypertensive action or guanethidine

and similar compounds. Avoid epinephrine in the treatment of phenothiazine-induced hypotension because phenothiazines may reverse epinephrine's action and therapy cause a further fall in blood pressure.

Adverse Effects: Adverse effects with different phenothiazines vary in type, frequent and mechanism of occurrence, i.e. some are dose related, while others involve individual patient sensitivity. Some adverse effects may be more likely to occur, or occur with greater intensity, in patients with special medical problems, e.g. patients with mitral insufficiency or pheochromocytoma have experienced severe hypotension following recommended doses of certain phenothiazines.

In general, members of the piperazine group of phenothiazines have more marked stimulating effects, are more likely to cause motor disorders associated with extrapyramidal reaction, particularly in children but are less likely to cause blood dyscrasias, hypotension, tachycardia, and drowsiness than the members of the other phenothiazine groups.

Not all of the following adverse reactions have been observed with every phenothiazine derivatives, but they have been reported with one or more and should be borne in mind when drugs of this class are administered. **Behavioral reaction:** oversedation; impaired psychomotor function; paradoxical effects, such as agitation, excitement, insomnia, bizarre dreams, aggravation of psychotic symptoms; and toxic confusional states. **CNS:** extrapyramidal reaction, including pseudoparkinsonism (with motor retardation, rigidity, mask like facies, pill rolling and other tremors, drooling, shuffling gait, etc.); dystonic reaction (including perioral spasms, and trismus, tics, torticollis, oculogyric crises, protrusion of the tongue, difficulty swallowing, carpopedal spasm and opisthotonos of the back muscles), and akathisia. Persistent dyskinesias resistant to treatment have been reported, particularly in elderly patients with previous brain damage. In addition, slowing of the EEG rhythm, disturbed body temperature and lowering of the convulsive threshold have occurred. Dizziness has been reported.

Tardive dyskinesia may appear in some patients on long-term antipsychotic therapy or may appear after drug therapy has been discontinued. The risk appears to be greater in elderly patients on high-dose therapy, especially females. The symptoms are persistent and in some patients appear to be irreversible.

The syndrome is characterized in rhythmical involuntary movements of the tongue, face, mouth or jaw (e.g. protrusion of tongue, puffing of cheeks, puckering of mouth, chewing movements). Sometimes these may be accompanied by involuntary movements of extremities.

There is no known effective treatment for tardive dyskinesia; antiparkinsonian agents usually do not alleviate the symptoms of this syndrome. All antipsychotic agents should be discontinued if these symptoms appear. Should it be necessary to reinstitute, or increase the dosage of the agent, or switch to a different antipsychotic agent, the syndrome may be masked. The physician may be able to reduce the risk of this syndrome by minimizing the unnecessary use of neuroleptics and reducing the dose or discontinuing the drug, if possible, when manifestations of this syndrome are recognized, particularly in patients over the age of 50. Fine vermicular movements of the tongue may be an early sign of the syndrome. If the medication is stopped at that time, the syndrome may not develop.

Autonomic nervous system: dry mouth, fainting, stuffy nose, photophobia, blurred vision, miosis. **Gastrointestinal:** anorexia, increased appetite, gastric irritation, constipation, nausea, vomiting, paralytic ileus. **Endocrine system:** altered libido, menstrual irregularities, lactation, false positive pregnancy tests, inhibition of ejaculation, gynecomastia, weight gain. **Skin:** itching, rash, hypertrophic papillae of the tongue, angioneurotic edema, erythema, allergic purpura, exfoliative dermatitis, contact dermatitis. **Cardiovascular effects:** hypotension, tachycardia, ECG changes (see precautions). **Blood dyscrasias:** agranulocytosis, leukopenia, granulocytopenia, eosinophilia, thrombocytopenia, anemia, aplastic anemia, pancytopenia. **Allergic reaction:** fever, laryngeal edema, angioneurotic edema, asthma. **Hepatotoxicity:** jaundice, biliary stasis. **Urinary disturbances:** retention, incontinence. **Abnormal pigmentation:** More recently, a peculiar skin-eye syndrome has been recognized as an adverse effect following long-term treatment with phenothiazines. This reaction is marked by progressive pigmentation of areas of skin or conjunctiva and/or discoloration of the exposed sclera and cornea. Opacities of the anterior lens and cornea described as irregular or stellate in shape have also been reported. Patients receiving higher doses of phenothiazines for prolonged periods should have periodic complete eye examinations. **Miscellaneous:** Unexpected and sudden deaths have been reported in hospitalized psychotic patients receiving phenothiazines. In some unexpected deaths, myocardial lesions have been observed. Previous brain damage or seizures may also be predisposing factors; high doses should be avoided in known seizure patients. Several patients have shown sudden exacerbations of psychotic behavior patterns shortly before death. Autopsy findings have also revealed acute fulminating pneumonia or pneumonitis and aspiration of gastric contents. The physician should therefore be alerted to the possible development of silent pneumonias. **Neuroleptic Malignant Syndrome:** As with other neuroleptic drugs, a symptom complex sometimes referred to as neuroleptic malignant syndrome (NMS) has been reported. Clinical features of NMS are hyperpyrexia, muscle rigidity, altered mental status (including catatonic signs), and evidence of autonomic instability (irregular pulse or blood pressure). Additional signs may include elevated CPK, myoglobinuria (rhabdomyolysis), and acute renal failure. NMS is potentially fatal, requires intensive symptomatic treatment and immediate discontinuation of neuroleptic treatment.

Overdose: Symptoms: Symptoms of minor overdosage include drowsiness, dizziness or muscular twitching. Gross overdosage may produce CNS depression, dystonia, torticollis, tremor, salivation, difficulty in swallowing and disturbances of gait.

Treatment: Treatment is essentially symptomatic and supportive. In adults, perform gastric lavage or, in children, induce emesis and if there is no immediate response, use gastric lavage. Centrally acting emetic drugs to induce emesis are ineffective due to the antiemetic effect of perphenazine. Stimulants that may cause convulsions (e.g., picrotoxin or pentylenetetrazol) should be avoided. Hypotension should be treated with levarterenol or phenylephrine. Epinephrine should not be used because it may cause further hypotension. Extrapyramidal symptoms may be treated with diphenhydramine or benztropine mesylate.

Route of Administration: Oral

Dosage: Must be adjusted for each patient according to the severity of the condition and the response obtained. The total daily dose in ambulatory patients ordinarily should not exceed 24mg. Severely disturbed hospitalized psychiatric patients or those with

resistant mental and emotional disorders may temporarily require more than 24mg daily, especially during early management. It is very important to employ the lowest effective dose since extrapyramidal symptoms increase in frequency and severity with increased dosage. For the rapid control of acute conditions, 5mg be injected intramuscularly. This dose may be repeated every 6 hours, but oral therapy should be instituted as soon as possible. Perphenazine should not be administered intravenously except during surgery to counteract retching, hiccups, and vomiting; this route of administration is not recommended in neuropsychiatric conditions. For control of nausea and vomiting, daily doses of 8 to 24mg may be given in divided amounts. Children: The lower range of the adult dosage may be used in children over 12. Dosage for children under 12 years of age has not been established to date. In nausea and vomiting 4 to 8mg once or twice daily may be given to control symptoms.

Product Description

White, round, biconvex, film-coated tablets, engraved '4' on one side, other side plain.

Pack Size:

PVC Blister of 500's and 1000's

Storage Conditions:

Store below 30°C

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